



Geographic differences in early-onset breast cancer incidence trends in the USA, 2001–2020, is it time for a geographic risk score?

Rebecca D. Kehm^{1,2,4} · Josephine M. Daaboul^{1,3} · Parisa Tehranifar^{1,2} · Mary Beth Terry^{1,2}

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Abstract

Purpose Breast cancer (BC) incidence is increasing in US women under 40, with variation across racial and ethnic groups. It is not yet known if incidence trends also vary by geography within the USA, which may inform whether place-based exposures contribute to BC risk in younger women.

Methods Using the US Cancer Statistics database, we analyzed age-adjusted BC incidence rates from 2001 to 2020 in women aged 25–39. We calculated the average annual percent change (AAPC) using Joinpoint regression and performed age-period-cohort analyses.

Results From 2001 to 2020, BC incidence in women under 40 increased by more than 0.50% per year in 21 states, while remaining stable or decreasing in the other states. Incidence was 32% higher in the five states with the highest rates compared to the five states with the lowest rates. The Western region had the highest rate of increase (AAPC = 0.76, 95% CI 0.56–0.96), despite having the lowest absolute incidence rate from 2001 to 2020. The Northeast had the highest absolute rate of BC among women under 40 and experienced a significant increase over time (AAPC = 0.59, 95% CI 0.36–0.82). The South was the only region where BC under 40 did not increase from 2001 to 2020.

Conclusion These findings support that BC incidence trends in US women under 40 vary by geography, and the range of state-specific risks was comparable in magnitude to other risk measures, such as polygenic risk scores. This suggests that incorporating place-based factors alongside established risk factors into risk prediction may improve our ability to identify groups of younger women at higher risk for early-onset BC.

Keywords Early-onset breast cancer · Cancer incidence trends · Young women · Geography · Disparities · Age-period-cohort analysis

Breast cancer (BC) incidence in younger women (also referred to as early-onset BC) has steadily increased over time in the USA and globally [1–5]. From 2004 to 2017, invasive BC incidence in this age group increased by 0.54%

annually, in women under age 40 with the incidence of distant stage disease growing even more rapidly at 3.54% per year in this age group [2]. This alarming increase in incidence cannot be explained by genetic factors, which evolve over much longer periods nor by changes in screening practices given that women under 40 years are below the recommended age for routine mammography screening [6]. These patterns suggest that other factors, like environmental or lifestyle-related exposures, particularly during critical early-life periods [7], may be contributing to the increase in early-onset BC incidence over time. Identifying these emerging risk factors is crucial for developing effective prevention and intervention strategies.

While the causes behind the rising incidence of early-onset BC are not yet fully understood, studying how trends vary across different population subgroups can offer valuable insights and help generate hypotheses for future research.

✉ Rebecca D. Kehm
rk2967@cumc.columbia.edu

¹ Department of Epidemiology, Mailman School of Public Health, Columbia University, New York, NY, USA

² Herbert Irving Comprehensive Cancer Center, Columbia University Medical Center, New York, NY, USA

³ Department of Epidemiology, Fielding School of Public Health, University of California Los Angeles, Los Angeles, CA, USA

⁴ Department of Epidemiology, Mailman School of Public Health, Columbia University, 722 W. 168th Street, Room 708, New York, NY 10032, USA

For example, studies have identified racial and ethnic (racial/ethnic) differences in early-onset BC incidence trends in the USA [3, 8]. Specifically, the incidence of distant stage BC in women aged 25–39 years increased by 3.5% per year from 1976 to 2009 in non-Hispanic Black (NHB) compared to 2.7%, 2.4%, and 1.9% for non-Hispanic White (NHW), Asian or Pacific Islander (API), and Hispanic women, respectively [3]. Less is known about whether trends in early-onset BC incidence also vary by geography within the USA. Exploring geographical differences could help generate hypotheses about how place-based factors—such as environmental exposures (e.g., air pollution, environmental carcinogens), social influences (e.g., cultural norms around breastfeeding, alcohol consumption, physical activity), built environment (e.g., access to transportation and healthcare facilities), and policies (e.g., health policies supporting physical activity or paid leave for breastfeeding)—might contribute to BC risk in young women [9].

The lack of comprehensive data on geographical variations in early-onset BC incidence in the USA is largely due to reliance on the Surveillance Epidemiology and End Results (SEER) database, which only captures data from a limited subset of states (ranging from 8 to 22 depending on the year) [10]. Among the few studies that have examined BC incidence trends by geography in the USA [11–13], only one used nationally representative data, but it focused solely on estrogen receptor-negative BC and did not analyze trends by age group [12]. To address these gaps, we analyzed 2010–2020 data from the US Cancer Statistics (USCS) database, which covers all 50 states. Our study focused on BC incidence trends among women under 40, a group in which population-wide screening is not recommended. Our analysis extends beyond previous studies by examining trends across different states, regions, and metropolitan versus non-metropolitan areas and by further stratifying by racial and ethnic groups. Additionally, we compared BC incidence trends in older age groups for which population-wide screening is recommended, providing a comprehensive view of how trends vary by location, age, screening recommendations, and population subgroup.

Methods

Data source

We utilized the USCS public use database [14] to obtain BC incidence rates, which were age adjusted to the 2000 US standard population. The database also provided 95% confidence intervals and standard errors for the rates. The USCS is a comprehensive, population-based cancer surveillance system that represents approximately 99% of the US population [15]. It combines data from two major sources: the

Surveillance, Epidemiology, and End Results (SEER) program and the Centers for Disease Control and Prevention's (CDC) National Program of Cancer Registries (NPCR) [15]. These cancer registries gather data on all newly diagnosed cancers reported by medical facilities, including hospitals, clinics, radiation therapy centers, freestanding surgical centers, and pathology labs [15]. This information is coded uniformly and compiled annually into a single dataset [15]. We obtained approval to access de-identified cancer incidence data from the USCS database through a Public Use Research Data Agreement with the CDC and the National Cancer Institute. The study was exempt from ethical review and adheres to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for reporting observational studies.

Study population

We included all invasive female BC cases diagnosed from 1/1/2001 through 12/31/2020 (all available years of data in the USCS database at the time of data extraction). Male BC cases were excluded due to limited data when stratified by age, geography, and race/ethnicity. Therefore, all BC incidence discussed here refers to female BC incidence. Using SEER*Stat software, we obtained annual age-adjusted BC incidence rates for the following age groups: 25–39, 40–54, 55–69, and 70–84 years. For each age group, we further stratified BC incidence rates by different geographical designations. This included stratification by state (plus the District of Columbia), region (Midwest, Northeast, South, West), and metropolitan or non-metropolitan designation. North Dakota, Vermont, and Wyoming were excluded from the state-specific analysis of the 25–39 years age group, because these states had multiple years of suppressed data due to small case counts. Regional classifications were based on US census definitions [16], and metropolitan designations were determined using Rural–Urban Continuum Codes from the US Department of Agriculture [17]. For each region, we obtained annual age-adjusted BC incidence further stratified by race/ethnicity (NHW, NHB, non-Hispanic API, Hispanic). Other racial/ethnic groups (e.g., American Indian and Alaska Native) were included in the overall analyses but excluded from race/ethnicity-specific analyses due to small sample sizes after further stratification by age group and geography. We also obtained early-onset BC incidence rates for each region stratified by estrogen receptor (ER) subtype (ER positive, ER negative, and ER unknown). However, data on ER subtypes were only available from 2004 to 2020.

Statistical analysis

We used SEER*Stat to calculate incidence rate ratios and 95% confidence intervals, applying the [18] modification for

CI estimation [18], to compare age-specific BC incidence rates across different geographic areas, racial and ethnic groups, and ER subtypes.

Joinpoint regression: We used joinpoint regression to estimate the average annual percent change (AAPC) in BC incidence rates from 2001 to 2020. This method fits a series of connected straight lines on a logarithmic scale to capture trends in the annual age-standardized rates [19, 20]. For the 25–39 age group, we applied a 3-year moving average to stabilize the AAPCs due to smaller case counts in certain subgroups. The models allowed for up to three joinpoints with at least two observations between each inflection point. The AAPC was calculated by taking a weighted average of the slope coefficients from the joinpoint regression line, with weights based on the length of each segment. To assess whether the AAPC was statistically different from zero ($p < 0.05$), we used a two-sided t test for models with zero joinpoints and a two-sided z test for models with one or more joinpoints. Rates were considered to be statistically significantly increasing or decreasing if $p < 0.05$; otherwise, they were classified as stable over the 2001–2020 period. The COVID-19 pandemic caused significant disruptions in cancer diagnoses in the US during 2020, leading to an estimated 13% lower incidence of all-site cancers than expected between March and December of that year [21]. Consequently, because joinpoint regression analysis is not suited for handling anomalies in data over just one year [22], we performed a sensitivity analysis that excluded the 2020 incidence rates from the joinpoint regression models. All analyses were performed using Joinpoint software version 4.9.1.0 (released April 2022) from the National Cancer Institute.

Age-period-cohort: To further explore whether age-specific BC incidence trends varied by region, we conducted an age-period-cohort analysis using the web tool (<https://analyzistools.nci.nih.gov/apc/>) developed by the National Cancer Institute's Division of Epidemiology and Genetics [23]. This method partitions the effects of age, period, and cohort on disease incidence or mortality rates using weighted least squares regression [23]. The issue of identifiability, or disentangling these three interrelated variables, is addressed by an algorithm that optimizes the estimation of each component independently. The age-period-cohort model helps in understanding how BC incidence rates change across different age groups (age effect), calendar periods (period effect), and birth cohorts (cohort effect).

We conducted a separate age-period-cohort analysis for each region of the USA. For each region, we divided the data into four five-year time periods (2001–2005, 2006–2010, 2011–2015, and 2016–2020) and 12 five-year age groups (25–29 to 80–84 years), spanning 15 five-year birth cohorts from 1921–1925 to 1991–1995. For each region, we estimated the net drift, which is the annual percent change in the overall expected age-adjusted rates

over time, reflecting combined period and cohort effects. We also estimated the local drift, which is the annual percent change in age-specific BC incidence rates over time, showing trends for individual age groups. Additionally, we estimated the period rate ratios (ratio of age-specific rates in each period relative to reference period of 2001–2005) and the birth cohort rate ratios (ratio of age-specific rates in each cohort relative to reference birth cohort of 1921–1925) for each region. All statistical tests were two-sided, and P values < 0.05 were considered statistically significant.

Results

From 2001 to 2020, the overall incidence of early-onset BC ranged from 28.6 per 100,000 in Wyoming to 40.8 per 100,000 in Connecticut. The five states with the highest early-onset BC incidence from 2001 to 2020 were Maryland, New York, New Jersey, Hawaii, and Connecticut. The average incidence rate in these five states was 1.32 times higher than the average rate in the five states with the lowest incidence rates (Idaho, North Dakota, Arizona, Utah, and Wyoming). For comparison, the difference in average incidence rates between the top and bottom 5 states was 1.32 for the 40–54 age group, 1.18 for the 55–69 age group, and 1.22 for the 70–84 age group (Supplementary Materials, Table S1). In all but six states, early-onset BC incidence rates were higher in 2011–2020 compared to 2001–2010 (Fig. 1).

When stratified by region, early-onset BC incidence rates ranged from 32.6 per 100,000 in the West to 38.7 per 100,000 in the Northeast (Fig. 2; Supplementary Materials Table S2). Across all four regions of the US, NHB women had the highest early-onset BC incidence, ranging from 39.3 per 100,000 in the West to 44.3 per 100,000 in the South. Hispanic women had the lowest early-onset BC incidence rates in all regions, ranging from 25.8 per 100,000 in the Midwest to 32.6 per 100,000 in the Northeast. In all four regions, the rate of ER-positive early-onset BC was more than twice as high as the rate of ER-negative early-onset BC (Fig. 3). The Northeast had the highest incidence rates for both ER-positive and ER-negative early-onset BC from 2004 to 2020. The South had the lowest rate of ER-positive early-onset BC, while the West had the lowest rate of ER-negative early-onset BC. The Northeast had a notably higher incidence of ER-positive BC cases compared to the South, with 5 more cases per 100,000 people. This difference was much greater than the increase in ER-negative cases in the Northeast compared to the West, which was only 1.4 cases per 100,000, indicating that regional disparities are more pronounced for ER-positive breast cancer. The South had more than double the rate of early-onset BC cases with unknown

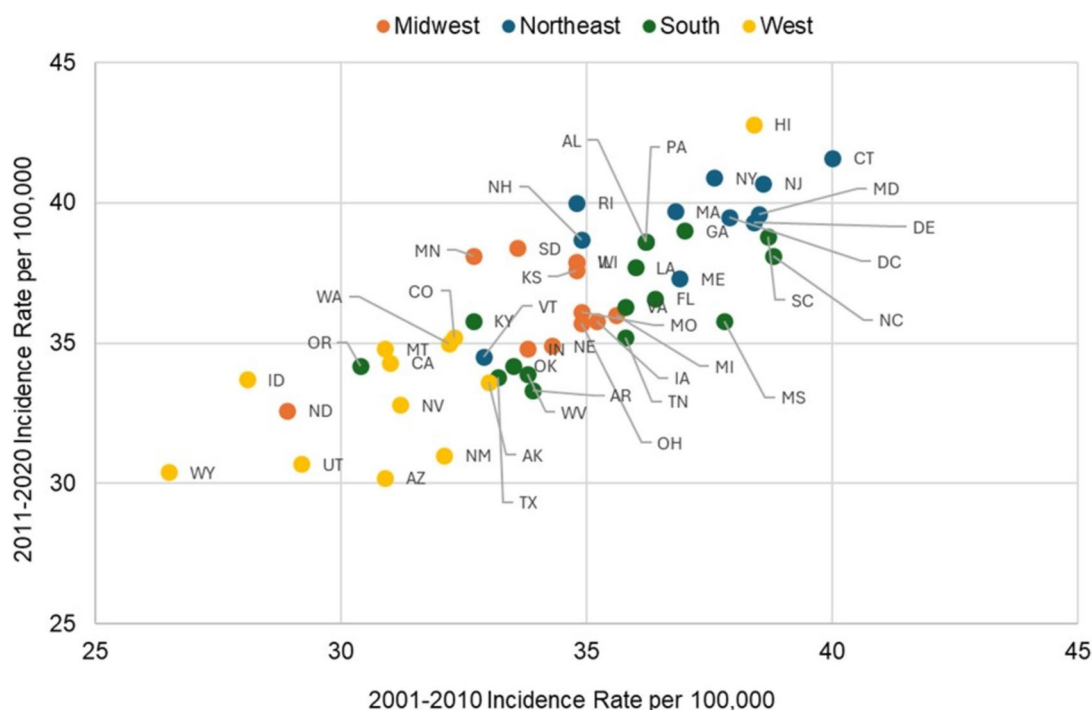


Fig. 1 Early-onset breast cancer incidence rates from 2001–2010 to 2011–2020 by state and region, United States Cancer Statistics Database. Early-onset breast cancer incidence rates include women diag-

nosed between ages 25 and 39 years. The rates are age-adjusted to the 2000 US standard population and include all racial and ethnic groups

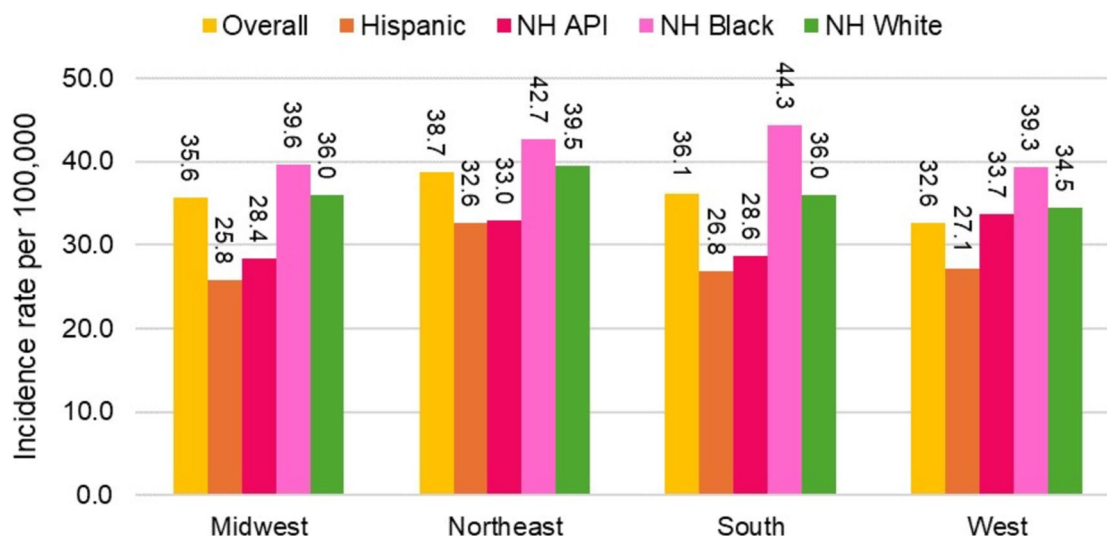


Fig. 2 Early-onset breast cancer incidence rates by region, race, and ethnicity, United States Cancer Statistics Database, 2001–2020. Early-onset breast cancer incidence rates include women diagnosed between ages 25 and 39 years. The rates are age adjusted to the 2000

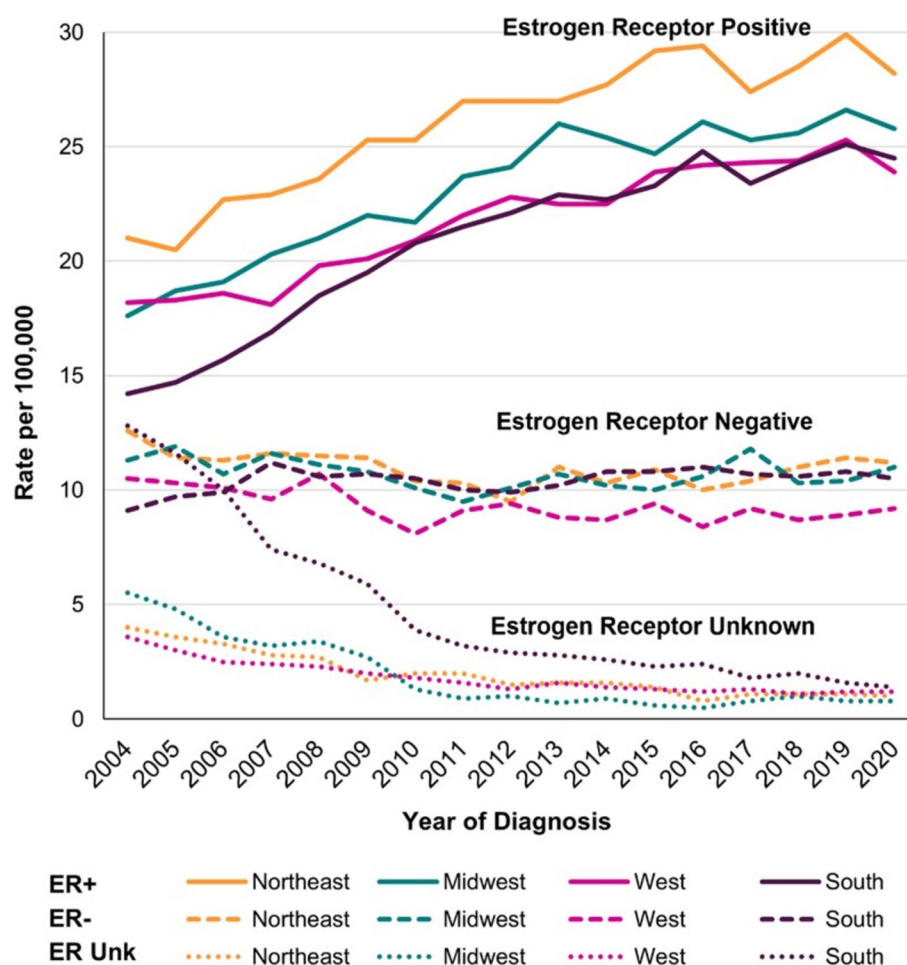
US standard population. Overall incidence rates includes all racial and ethnic groups, including American Indian and Alaska Natives. *API* Asian and Pacific Islander; *NH* non-Hispanic

ER status, largely due to substantially missing data in earlier years of the study (e.g., 12.8% missing in 2004 compared to 1.4% missing 2020).

Joinpoint regression results

When stratified by state, early-onset BC incidence increased by more than 0.50% per year from 2001 to 2020 in 21 states,

Fig. 3 Early-onset breast cancer incidence rates by estrogen receptor status, United States Cancer Statistics Database, 2004–2020. Early-onset breast cancer incidence rates include women diagnosed between ages 25 and 39 years. The rates are age adjusted to the 2000 US standard population and include all racial and ethnic groups. *ER+* estrogen receptor positive, *ER-* estrogen receptor negative, *ER Unk* estrogen receptor unknown



Region	ER Positive		ER Negative		ER Unknown	
	Incidence Rate	Incidence Rate Ratio	Incidence Rate	Incidence Rate Ratio	Incidence Rate	Incidence Rate Ratio
Northeast	26.0	ref.	11.0	ref.	2.0	ref.
Midwest	23.1	0.89 (0.87, 0.91)	10.7	0.98 (0.95, 1.00)	1.9	0.97 (0.91, 1.04)
South	21.0	0.81 (0.79, 0.82)	10.4	0.95 (0.93, 0.97)	4.7	2.37 (2.25, 2.50)
West	21.9	0.84 (0.83, 0.86)	9.3	0.85 (0.82, 0.87)	1.8	0.90 (0.84, 0.96)

although the increase was only statistically significant in 12 states. Early-onset BC incidence statistically significantly decreased in one state (Mississippi) and remained stable in the remaining states (Fig. 4). Of the 12 states with a statistically significant increase in early-onset BC incidence, three were in the Midwest (Illinois, Minnesota, Wisconsin), four were in the Northeast (Massachusetts, New Hampshire, New York, Pennsylvania), two were in the South (Delaware, Louisiana), and three were in the West (California, Colorado, Oregon). State-specific AAPCs for the older age groups are provided in the Supplementary Materials (Table S3).

In the USA overall, early-onset BC incidence increased by 0.42% per year (95% confidence interval (CI) 0.26, 0.58) from 2001 to 2020. For comparison, BC incidence in the 40–54 age group increased by 0.28% per year (95%

CI 0.10, 0.45), while rates remained stable in the 55–69 age group over the same period. In the 70–84 age group, BC incidence decreased by 0.43% per year (95% CI -0.83, -0.02) from 2001 to 2020; however, when the 2020 incidence rate was excluded, the decrease was smaller and not statistically significant (2001–2019 AAPC = -0.20, 95% CI -0.55, 0.15). Conclusions about the overall trends in BC incidence rates did not change for the younger three age groups after excluding 2020 data (Supplementary Materials, Table S4).

When states were grouped into regions, early-onset BC incidence statistically significantly increase from 2001 to 2020 in all regions except the South, where the rate remained stable (AAPC = 0.17, 95% CI -0.03, 0.38). Statistically significant increases in AAPCs ranged from 0.52% (95% CI 0.30, 0.74) in the Midwest to 0.76% (95%

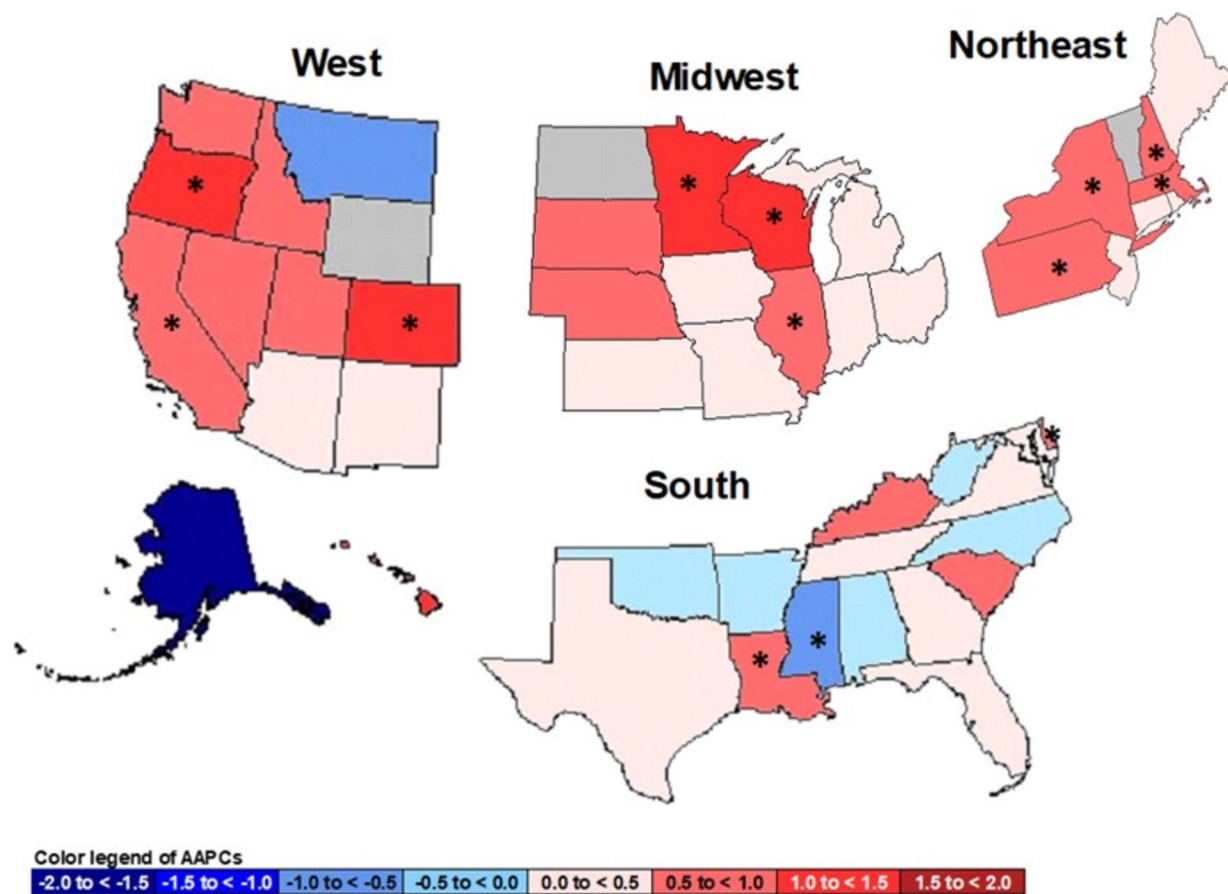


Fig. 4 Average annual percent change in early-onset breast cancer incidence by state, United States Cancer Statistics Database, 2001–2020. Average annual percent changes (AAPCs) were estimated using 3-year moving averages to stabilize rates over time. States with a statistically significant AAPC are denoted with an asterisk (i.e., p

value < 0.05 testing whether AAPC different from 0%). The AAPCs for Indiana and Nevada were calculated using 2001–2019 incidence data. The AAPC for Mississippi was calculated using 2003–2020 incidence data. Refer to the Supplementary Materials, eTable 1, for the AAPCs and 95% confidence intervals for all 50 states

CI0.56, 0.96) in the West (Fig. 5). When further stratified by metropolitan designation (see Supplementary Materials, Tables S5 for the case counts), early-onset BC incidence statistically significantly increased in metropolitan areas across all four regions, with AAPCs ranging from 0.16% (95% CI0.02, 0.30) in the metropolitan South to 0.75% (95% CI0.55, 0.95) in the metropolitan West. In non-metropolitan areas, statistically significant increases were observed only in the Northeast (AAPC = 0.63, 95% CI0.29, 0.97) and the West (AAPC = 1.17, 95% CI0.09, 2.27), with no significant changes in the Midwest or South.

When further stratified by race and ethnicity, NHW women were the only group to experience a statistically significant increase in early-onset BC incidence across all four regions of the US, with AAPCs ranging from 0.37% (95% CI0.21, 0.54) in the South to 0.82% (95% CI0.68, 0.96) in the West. Non-Hispanic API women experienced an increase in early-onset BC incidence in the Midwest (AAPC = 1.38, 95% CI1.03, 1.74) and West (AAPC = 0.76,

95% CI0.52, 1.00). NHB women only experienced a statistically significant increase in early-onset BC incidence in the Northeast (AAPC = 1.08, 95% CI0.21, 1.97), while Hispanic women only experienced an increase in the West (AAPC = 0.93, 95% CI0.10, 1.76). Regional differences in BC incidence rates were also observed in the older age groups (Supplementary Materials, Table S4).

Age-period-cohort analysis results

From 2001 to 2020, BC incidence showed a statistically significant positive net drift in the Midwest (0.36, 95% CI0.24, 0.48), Northeast (0.39, 95% CI0.26, 0.53), and South (0.18, 95% CI0.07, 0.29) (Fig. 6; Supplementary Materials, Table S6). The West was the only region without a statistically significant net drift or period effects (i.e., no significant differences in rate ratios when comparing the later periods to 2001–2005). Notably, the South was the only US region where no statistically significant positive

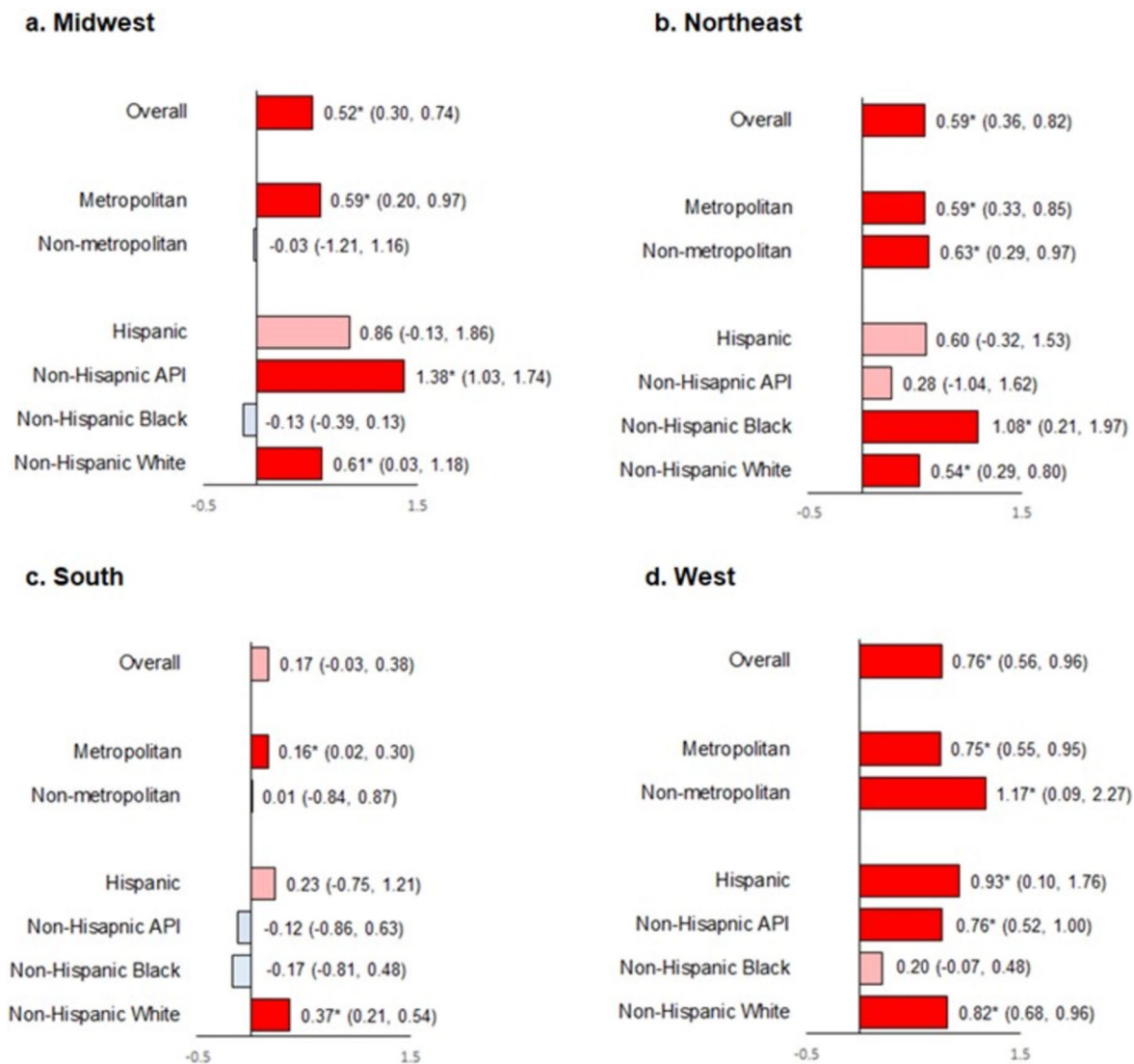


Fig. 5 Average annual percent change in early-onset breast cancer incidence by region, metropolitan designation, and race and ethnicity, United States Cancer Statistics Database, 2001–2020. Average annual percent changes (AAPCs) were estimated using 3-year moving aver-

age to stabilize rates over time. The 95% confidence interval for each AAPC is provided in parentheses. Statistically significant AAPCs (p value < 0.05 testing whether AAPC different from 0%) are denoted with an asterisk

local drifts were observed in any age group between 25 and 39 years, aligning with the results from the joinpoint regression analysis.

Discussion

Using data from cancer registries in all 50 states, this study provides a comprehensive analysis of recent geographical trends in early-onset BC incidence in the USA. Consistent with previous research [1–3], we found that early-onset BC incidence has increased in recent decades in the USA overall, increasing by nearly 0.50% per year since 2001. However, our state-level analysis revealed substantial variability

in early-onset BC incidence trends across states from 2001 to 2020. Although early-onset BC incidence increased by more than 0.50% in 21 states, the increase was statistically significant in only 12 of them. In the remaining states, early-onset BC incidence either remained stable or declined from 2001 to 2020. We found that the average rate of early-onset BC in the five states with the highest incidence was 32% higher than in the five states with the lowest incidence. This difference is comparable to the risk gap between individuals in the highest and lowest percentiles of absolute risk, as determined by polygenic risk scores [24], suggesting that geography may play a significant role in risk stratification. This difference was also as large or even larger than the differences observed in the three older age groups. This is

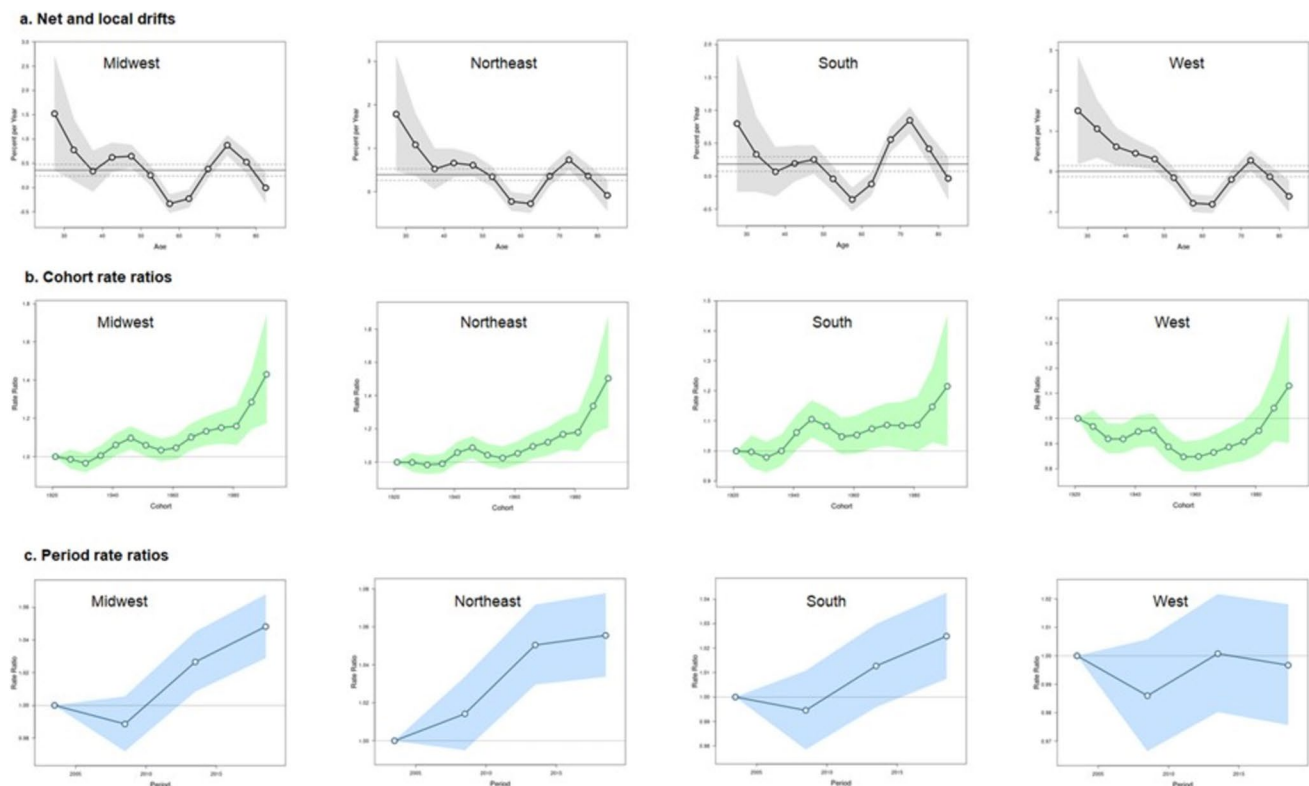


Figure 3. Age, period, and cohort effects for breast cancer incidence rates over time by region, United States Cancer Statistics Database, 2001–2020

Fig. 6 Age, period, and cohort effects for breast cancer incidence rates over time by region, United States Cancer Statistics Database, 2001–2020. **a** Net drift, represented by a solid horizontal line on the graph, represents the annual percent change of the expected age-adjusted rates over time. Local drifts represent the annual percent

change of the expected age-specific rates over time for each 5-year age interval. **b** Cohort rate ratios represent the ratio of age-specific rates in each cohort relative to the reference cohort (1921–1925). **c** Period rate ratios represent the ratio of age-specific rates in each period relative to the reference period (2001–2005)

notable because state-level differences in screening practices are unlikely to contribute to disparities in early-onset BC incidence, whereas such differences are more likely to influence incidence rates in the older age groups [6]. These data highlight the potentially significant impact of environmental and other place-based risk factors on early-onset BC risk. Given the latency of tumorigenesis, this may include exposures occurring in early-life windows of susceptibility (e.g., prenatal and adolescent periods) [7].

When we examined trends by region, we found that early-onset BC increased from 2001 to 2020 in every region except the South. However, it is important to note that there is likely to be variation in age-specific BC incidence rates within regions. For example, Hawaii had substantially higher early-onset BC incidence rates compared to other states in the West, while North Dakota had lower rates than other states in the Midwest. Additionally, when we further stratified the trends by metropolitan designation, we found that early-onset BC incidence rates increased in metropolitan areas across all US regions between 2001 and 2020. However, in non-metropolitan areas, increases were only observed in the

Northeast and West. The age-period-cohort analysis revealed additional regional differences in age-specific BC incidence trends over time. For example, the South was the only region without significant positive local drifts in the age groups under 40, while the West was the only region without a statistically significant positive net drift (across all age groups) from 2001 to 2020. While this study focused primarily on women under 40, we also identified geographical variations in incidence trends across older age groups. These findings provide direction for hypotheses about the underlying causes of changing BC incidence rates in the USA. For example, it may be important for future etiological studies to consider risk factors that are lower in prevalence in non-metropolitan areas of the Midwest and South compared to other areas of the USA. This may include certain issues around quality of and access to care, environmental exposures (e.g., air pollution or endocrine disrupting chemicals) [25], lifestyle factors, and reproductive factors. For example, birth rates and maternal age are shown to vary across US states [26], which may reflect differences in certain state policies, such as those governing contraceptive insurance mandates [27].

However, our previous research suggests that changes in fertility rates alone are unlikely to fully account for increasing early-onset BC incidence rates in the USA or globally [4, 28]. Therefore, it is important to investigate other risk factors that vary by location in relation to the increasing rates of early-onset BC incidence. This should include factors like alcohol consumption, an established risk factor for BC [29, 30], which is shown to vary across states and is influenced by state alcohol policies [31].

We also observed variations in early-onset BC incidence when analyzing the data by race/ethnicity. Consistent with previous research [3, 32], we found that NHB women had the highest incidence of early-onset BC. This was true across all geographic regions, with NHB women in the South experiencing the highest incidence overall. NHW women were the only racial/ethnic group to experience a statistically significant increase in early-onset BC incidence from 2001 to 2020 across all four US regions, including in the South, where there was not an overall statistically significant increase. No US region showed a significant rise in early-onset BC incidence across all four racial/ethnic groups. Moreover, each region had a different group with the highest rate of increase: non-Hispanic Asian/Pacific Islanders in the Midwest, non-Hispanic Black women in the Northeast, non-Hispanic White women in the South, and Hispanic women in the West. These findings demonstrate why stratification by only one factor at a time (e.g., geography or race/ethnicity) may conceal important subgroup differences in BC incidence trends in the USA. These findings support how the intersection of structural inequalities embedded in certain place-based factors and racial/ethnic marginalization, like historical mortgage lending practices [33], may disproportionately affect some subgroups of the population more than others. For example, NHB women in states historically governed by Jim Crow laws have been found to have a higher risk for estrogen receptor-negative BC compared to NHB women in states without such laws. However, this association does not hold for NHW women [34]. Multilevel interventions and policies that address factors at the local, state, and national levels are thus likely needed to address disparities in BC incidence in the USA [35, 36].

Beyond identifying variations in temporal trends in BC incidence, our findings highlight the importance of considering the absolute rates of BC across diverse populations. For example, while the West had the highest relative rate of increase (i.e., AAPC) in early-onset BC incidence from 2001 to 2020, it still had the lowest absolute rate of early-onset BC incidence. Additionally, while NHB women only experienced a statistically significant increase in early-onset BC incidence in the Northeast, they still had the highest absolute incidence for this age group across all four regions of the USA from 2001 to 2020. These examples highlight the importance of evaluating both the

overall burden of BC incidence at a specific point in time and the changing patterns over time, especially when it comes to identifying and addressing disparities.

Study strengths and limitations

This study has several strengths, including the use of cancer registry data from all 50 states in the USA. This is also one of the first studies to evaluate BC incidence trends stratified by age, geography, and race/ethnicity simultaneously, which allowed us to identify more precise subgroups of the population that are disproportionately burdened by BC incidence in the USA. We were limited, however, by the number of years we could use to assess trends given that the USCS public use database only includes data going back to 2001. Therefore, we were not able to consider historical trends in age-specific cancer incidence rates, as we have done in prior work [1, 28, 37]. Historical trends may be important for identifying exposures that have changed more gradually over a long period of time. Focusing on more recent trends in BC incidence rates is important for understanding and addressing (e.g., informing state and regional cancer policies) contemporary BC disparities. Another limitation of this study is that we may have been underpowered to identify significant trends in BC incidence within certain subgroups, such as in states with smaller populations and non-metropolitan areas. To address this, we used 3-year moving averages to stabilize rates in the youngest age group. However, further studies with longer observation periods may be able to identify trends in smaller geographic areas that were not revealed in this study. The issue of having small case counts within certain subgroups also prevented us from further analyzing breast cancer incidence trends by molecular subtype, as this would have led to very small numbers for some subgroups. Additionally, we did not analyze BC incidence trends at more detailed geographic levels, such as counties or cities, so we could not assess variations within states. Nevertheless, analyzing trends at the state or regional level can provide valuable insights into how state-level policies and other regional factors may influence age-specific BC incidence rates over time. We were also limited in that the USCS database does not include data on risk factors and, therefore, we cannot make inferences about the underlying drivers of changing BC rates. Finally, we defined early-onset BC as occurring before age 40 to minimize the influence of screening on our findings of incidence trends. However, other studies have used different age cut-offs, such as before age 45 [38], making direct comparisons with our results challenging. Nevertheless, our study provides important insights into the increase in BC incidence among women who are not currently recommended for routine screening.

Conclusion

This study adds to the evidence that early-onset BC incidence is increasing in the US overall, although rates have remained stable in many states over the past few decades. Substantial differences in these trends were observed when we analyzed the data by state, region, metropolitan status, and race/ethnicity. These findings suggest that incorporating place-based factors, along with other established risk factors, into risk prediction may enhance our ability to identify groups of younger women who are at higher risk for early-onset BC. This, in turn, can inform targeted strategies for risk reduction, screening, and broader policy and community-based efforts to mitigate the rising rates. This is supported by our observation that the variation in early-onset BC incidence across states is similar to the variation in risk seen with polygenic risk scores. Further research is needed to understand how place-based factors, along with other known and emerging risk factors, contribute to changing age-specific BC incidence across different US subgroups.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10552-025-01968-7>.

Author contributions R.D.K contributed to the conception and design of the study; the acquisition, analysis, and interpretation of data; and the drafting and revising of the manuscript. J.M.D contributed to the acquisition, analysis, and interpretation of data, as well as the drafting and revising of the manuscript. P.T contributed to the interpretation of data and revising of the manuscript. M.B.T contributed to the conception and design of the study, interpretation of the data, and revising of the manuscript.

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Data availability The dataset used and analyzed during this study are available from the US Cancer Statistics Public Use Database (<https://www.cdc.gov/united-states-cancer-statistics/public-use/index.html>).

Declarations

Competing interests The authors declare no competing interests.

Ethical approval This study was performed in line with the principles of the Declaration of Helsinki. We were approved for data usage from a US Cancer Statistics Public Use Research Data Agreement with the Centers for Disease Control and National Cancer Institute to access the de-identified cancer incidence data in the USCS database. The study was exempt from ethical review.

Consent to participate Not applicable.

Consent for publications Not applicable.

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